

Original Articles

Response of preferential flow to soil – root – rock fragment system in karst rocky desertification areas

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ARTICLE INFO

Keywords:

Preferential flow
 Karst rocky desertification
 Rock fragment
 Root system
 Soil property

ABSTRACT

The response of preferential flow to the soil – root – rock fragment systems is essential for understanding the eco-hydrological processes in karst rocky desertification (KRD) areas. In this study, we investigated the soil properties, root, and rock fragment distribution characteristics and preferential flow in this system. Results show that there are differences in the development of preferential flow under five grades of rocky desertification: light rocky desertification (LRD) > potential rocky desertification (PRD) > non-rocky desertification (NRD) > moderate rocky desertification (MRD) > severe rocky desertification (SRD). Different degrees of rocky desertification have facilitating or inhibiting effects on preferential flow. LRD and PRD promote preferential flow with ratio of 6.6 %–9.6 %, while MRD and SRD inhibit preferential flow with ratio of 4.2 %–6.7 %. The preferential flow is simultaneously influenced by the components of soil – root – rock fragment systems. Root system indicators (i. e., root biomass, root bulk density, and root length density) generally decrease as grades of rocky desertification increase. The surface outcrops density, rock fragment density, and volume fraction tend to increase with grades of rocky desertification. A principal components analysis found that preferential flow of MRD and SRD is highly influenced by rock fragments, while LRD is more driven by root systems and soil properties. PRD is averagely influenced by the combination. These findings suggest that appropriate plant and rock fragment ratios can effectively avoid water loss in the form of preferential flow and more attention should be paid to the ecological problems of LRD and PRD areas.

1. Introduction

Land degradation is a substantial reduction in the sustainability of ecosystem services (Reynolds et al., 2007). One of the extreme forms of land degradation is rock desertification. The main cause of regional poverty in karst rocky desertification (KRD) areas is frequent human activities (Guo et al., 2021; Zheng et al., 2023). Besides, natural factors such as seasonal droughts exacerbate the occurrence of rocky desertification, creating a vicious circle of poor ecological quality and regional economic poverty in KRD (Zhang et al., 2016; Zheng et al., 2023). Seasonal drought may cause vegetation destruction, soil erosion, and deterioration of land quality, and even extreme irreversible degradation of ecosystem (Zhao et al., 2021).

Among the ecosystem degradation problems of KRD, the most urgent need to be concerned is the water loss in epikarst. The subsurface hydrological processes are influenced by ground cover characteristics, soil – rock distribution patterns, soil generation and weathering, and plant

root distributions in the epikarst zone (Dasgupta et al., 2006; Kogovsek and Petric, 2014). The response of water infiltration to the soil – root – rock fragment systems directly affects the health and balance of rocky desertification ecosystems (Liu et al., 2022). As a critical component of the epikarst system, soil materials, plant root networks, and rock fragments are the direct media for vertical water infiltration (Li et al., 2020). Water flow patterns of vadose zone include uniform flow and preferential flow. The preferential flow is an inhomogeneous flow of rapidly infiltrating water (Beven and Germann, 1982), with 70 %–85 % of soil water flow associated with preferential flow (Hlavacikova et al., 2019). Thus, revealing the response of soil, root, and rock fragment to subsurface hydrological processes plays a vital role in reflecting the dynamics of human-earth relations in the region and the restoration of KRD.

Soil particles are the basic units of soil structure and carriers of water infiltration movement. Soil physical properties alter the soil water infiltration characteristics and indirectly affect the development of

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preferential flow paths (Yan et al., 2022; Guo et al., 2022). Soil texture has a dual control on porosity and pore connectivity. Preferential flow typically occurs through macropore spaces with adequate porous connectivity (Tang et al., 2020; Yu et al., 2023a). Typical soils of carbonate-derived laterite together with localised roots and rock fragments form special underground karst pipes (Collon et al., 2017; Li et al., 2020).

Rock fragment refers to particles larger than 2 mm in diameter according to the USDA (Schoeneberger et al., 2012). Rock fragment contributes to the formation of macropore spaces between rock fragment and rock fragment, and between rock fragment and fine soil interfaces. These macropore phases promote water infiltration and increase soil capacity, thereby weakening hydrological processes such as surface runoff and reducing soil erosion (Huang et al., 2023; Sekucia et al., 2020). Four patterns of rock fragments are generally found in soils, i.e., resting on the soil surface, partially embedded into topsoil, completely embedded into topsoil, and completely embedded into deep soil (Zhang et al., 2016). Surface outcrops (i.e., the first two patterns of rock fragments) directly impact surface runoff processes. Guo et al. (2010) calculated that soil erosion and runoff are reduced ranging from 0 % to 40 %, and that suitable rock fragments can protect the surface from denudation. However, soil erosion occurs when the percentage of exposed rock fragments cover is higher than 60 % (Zavala et al., 2010). The latter two patterns of rock fragments are described as subsurface rock fragments. Studies have shown that the formation of fissures or deep holes characterized by vortices at the soil-rock interface (Backnäs et al., 2012; Huang et al., 2023), which is considered as the evidence that subsurface rock fragments can promote preferential flow surrounded by high concentrations of solute. But rock fragment content does not only promote preferential flow in one direction. When rock fragments reach a certain threshold, they obstruct the contact of water with the soil matrix, thus inhibiting preferential flow (Peng et al., 2014; Hou et al., 2023). Furthermore, soil particles at the soil-rock interfaces and in fissures is highly susceptible to leaking downward in the form of soil creep (Cen et al., 2023). Hence, in KRD areas, where rock fragment is a widely distributed medium of contact with the soil matrix, rock fragment responds to preferential flow most directly.

In the root-rock fragment system, the effect of rock fragments on root growth is positive, as the roots that grow with rock fragments usually tend to be well developed (Zhang et al., 2017). Root biomass and cumulative root bulk density determine the size of the root system (Figueroa-Bustos et al., 2018), which is due to the fact that root hairs significantly increase the surface area and root length of the plant root system (Glyn et al., 2016; Haling et al., 2013). In general, macroscopic porosity is markedly higher near the root zone than in the non-root zone (Yue et al., 2021). Moreover, secreted organic acids by the root system improves soil penetration and root-soil contact (Zhu et al., 2021). In KRD areas, plant roots tend to tolerate the stress of desertification by reducing root diameter and contact area with the soil (Yu et al., 2022). Both vertical and lateral water flow along the root can contribute to water redistribution (Wu et al., 2021), and Luo et al., (2022a) revealed that root-solute interactions are positively correlated with soil water content. Mair et al. (2022) considered the hydraulic characteristics of soils as a result of root-oriented preferential flow. They proposed a model for water infiltration with preferential flow oriented by plant roots. Zhang et al. (2021) derived that preferential flow promoted nutrients in the soil and enhanced the root-soil system.

Roots and rock fragments are included in the laterite red soil as the primary carriers of presence in the regolith and constitute the soil – root – rock fragment system. These features determine the erosion pattern of superimposed surface and subsurface loss in karst (Hou et al., 2023; Cen et al., 2023). Therefore, either medium in soil – root – rock fragment systems has impacts on the movement of water in the soil. Preferential flow, as a small-scale hydrological movement, will also affect larger-scale hydrological movement in the epikarst. However, the mechanism of preferential flow responding to this system under different grades of rocky desertification is unclear.

In this paper, we present a comprehensive evaluation system derived from classified identification analysis and traditional dye-stained tracer methods under different grades of rocky desertification. We target to (i) analyze the characteristics of soils, roots, and rock fragments under different grades of rocky desertification, (ii) assess the impact of root systems and rock fragments on preferential flow, and (iii) establish a comprehensive quantitative index of preferential flow in epikarst. The study of the coupled processes of soil, roots, rock fragments, and water have an essential role in the restoration and management of KRD.

2. Materials and methods

2.1. Study area

Our study was carried out within the Jiangshui County (102°54'00"~102°54'55"E, 23°36'50"~23°37'30"N), Yunnan Province of Southwest Region, China. Jiangshui County is located in the combination of the Eastern Yunnan Plateau and the Hengduan Mountains of Western Yunnan, driven by geological and climatic factors, and belongs to the comprehensive management area of KRD in the karst graben basin in southwest China (Cao et al., 2008; Xiao et al., 2020). Carbonate laterite is the dominant soil type in this area. The study area has a subtropical monsoon climate, with annual average temperature of 19.8 °C and precipitation of 805 mm. Precipitation is mainly concentrated between May and October due to the monsoon climate.

The classification of different rocky desertification is based on the 2016 survey data of the National Forestry and Grassland Administration (NFGA), the Technical Provisions for Monitoring Rocky Desertification in Karst Areas of China, and field surveys and references (Bai et al., 2011; Jiang et al., 2014). The rock fragment exposure, rock fragment content, average height of vegetation, vegetation coverage, and soil thickness are assigned values to categorize rocky desertification into five degrees (Table 1).

Five standard sites (20 m × 20 m) were chosen for the degree of rocky desertification (Fig. 1), namely non-rocky desertification (NRD), potential rocky desertification (PRD), light rocky desertification (LRD), moderate rocky desertification (MRD), and severe rocky desertification (SRD). The topographic conditions are similar between the different rocky desertification sites, and the bedrock is dominated by dolomite or limestone developed from carbonate rocks. In MRD and SRD sites, rock outcrops are widely distributed, and plants are growing along the rock crevices despite the high vegetation cover (Fig. 1). The vegetation cover, rock fragments exposure, and regolith features of the standard sites are shown in Table 2.

2.2. Samples collection

2.2.1. Soil sample collection and physical property measurement

The five-point sampling method randomly selected five points in different grades rocky desertification sites, while soil samples were collected. Samples were gathered in layers of 0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm, mixed thoroughly, and taken back to the laboratory. The soil samples were naturally air-dried and passed through a 2-mm sieve to remove impurities carefully. Soil particle size was determined by laser diffraction (Malvern Laser Particle Analyzer MS-3000). Soil particle size classification is based on the USDA (Schoeneberger et al., 2012) of clay (<0.002 mm), silt (0.002 – 0.05 mm), and sand (0.05–2 mm). The calculation of natural water content, total porosity, and capillary porosity is based on Dexter et al., (1988).

2.2.2. Roots collection and treatment

Root samples were collected by root bur in each soil layer at 10 cm intervals. The number of roots was low until the collection reached a depth of 40 cm. Root sampling points were chosen at 50 cm away from the soil sampling points. Ten points were randomly selected for each degree of rocky desertification. The soil column was taken out from the

Table 1
Criteria for classifying rocky desertification.

Rock fragment exposure		Rock fragment content		Average height of vegetation		Vegetation coverage		Soil thickness	
Value	Score	Value	Score	Value	Score	Value	Score	Value	Score
20 %~29 %	20	5 %~10 %	18	≥ 81 cm	2	50 %~69 %	2	≥40 cm	2
30 %~39 %	26	11 %~20 %	26	51 ~ 80 cm	4	30 %~49 %	8	20 ~ 39 cm	6
40 %~49 %	32	21 %~30 %	32	31 ~ 50 cm	8	20 %~29 %	14	10 ~ 19 cm	8
50 %~59 %	38	31 %~40 %	38	16 ~ 30 cm	10	10 %~19 %	20	<10 cm	10
≥60 %	44	≥41 %	44	≤15 cm	15	<10 %	26	—	—

Note. Three plots (30 m × 30 m) are randomly selected to determine the degree of rocky desertification based on the sum of the average scores of five assessment indicators. The specific criteria are as follows: no rocky desertification (NRD), sum of the scores ≤ 25; potential rocky desertification (PRD), sum of the scores ranging from 26 to 39; light rocky desertification (LRD), sum of the scores ranging from 40 to 53; moderate rocky desertification (MRD), sum of the scores ranging from 54 to 67; and severe rocky desertification (SRD), sum of the scores ≥ 68.

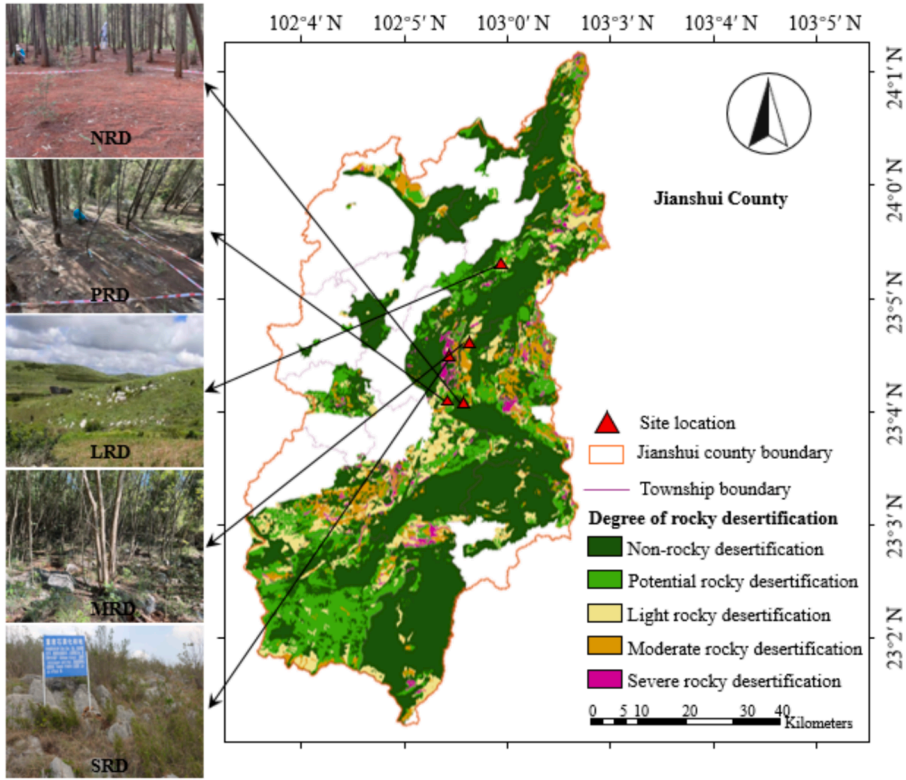


Fig. 1. Jianshui County different degrees of KRD classification.

Table 2
Characteristics of sites with different grades of rocky desertification.

Degree of rocky desertification	Slope gradient (°)	Elevation (m)	Integrated vegetation cover (%)	Average soil thickness (cm)	Average rock fragments exposure (%)	Regolith features	Vegetation
Non-rocky desertification (NRD)	13	1363.6	55	48	0	Limestone-dolomite-dolomitic limestone	Tall trees dominate, mainly <i>Cupressus duclouxiana</i> , almost no herbaceous covered
Potential rocky desertification (PRD)	11	1511.0	60	40	25	Dolomitic limestone	<i>Pinus yunnanensis</i> , <i>Cupressus duclouxiana</i> , sparse herbaceous covered
Light rocky desertification (LRD)	15	1376.7	70	40	60	Dolomitic limestone	Herbs dominate, mainly <i>Imperata cylindrica</i> , <i>Heteropogon contortus</i>
Moderate rocky desertification (MRD)	10	1394.0	70	35	70	Limestone-dolomite-aleurolite	Shrubs and herbs dominate, mainly <i>Agave americana</i> , <i>Bidens Pilosa</i> , <i>Ligustrum lucidum</i>
Severe rocky desertification (SRD)	13	1414.4	70	35	85	Limestone-biochip limestone-dolomite	Shrubs dominate, mainly <i>Damnacanthus indicus</i> , <i>Albizia kalkora</i>

root bur, and the soil around the root system was gently removed. The biomass of the plant roots is the weight ratio when dried to a constant weight to the sample volume. Then the residual soil attached to roots was cleaned with running water and wiped dry for the following analysis. The pre-treated roots were scanned with V850 Pro EPSON Perfection professional scanner (Long Beach, CA, USA) and analyzed with WinRHIZO software (2020a, Regent Instruments Inc., Quebec, QC, Canada) to extract the root bulk density and root length density under 1200 dpi.

2.2.3. Outcrops and rock fragments calculation

Outcrops were measured directly from the Feret diameter to estimate outcrop density. As some of the larger subsurface rock fragments are difficult to collect and excavate, the subsurface rock fragments were determined by images of the soil profile to estimate the number density and volume fraction of rock fragments of different diameter sizes (Sahagian and Proussevitch, 1998; Shea et al., 2010; Jiang et al., 2021; Hou et al., 2023). The outcrops and rock fragments were graded into gravel (2–76 mm), cobbles (76–250 mm), stones (250–600 mm), and boulders (>600 mm) according to the USDA (Schoeneberger et al., 2012).

2.3. Blue dye experiment

A total of 15 (5 grades of rocky desertification $\times$ 3 repeats) 60 $\times$ 60 cm plots were stained in Sept–Oct 2022. If the randomly selected plots encountered exposed bedrock with a diameter higher than 60 cm, staining profiles containing the complete soil-rock interface were excavated in order not to affect and destroy the stained profile images. The final staining profiles were removed 5 cm from both the left and right sides, with an image width of 50 cm. The aim was to avoid the influence of staining edge effects on the experimental results and to ensure consistency in data analysis. The experiment was conducted on days when no rainfall had fallen the day before. Each plot was sprayed with 7.2 L of 4 g/L Brilliant Blue FCF (C.I. Food Blue 2) to visualize the preferential flow paths. The solution volume was based on the events that more than 65 % of the rainfall in this region exceeded 20 mm from 2009 to 2018. Plastic films were covered over the plots after the staining to prevent the effect of evaporation on the results. Vertical soil profile was excavated 48 h after staining and 5 cm intervals as a staining level. Excess soil in each profile was carefully removed with a small spatula to avoid roots and rock fragments. The rock fragments embedded in soil profile were exposed by brushes to facilitate identification and collection of profile rock fragments. Image data were collected with a digital camera (Canon EOS 60D, Canon, Japan). Preferential flow ratio (P_F) is the ratio of the stained area of the preferential flow area to the entire vertical profile (van Schaik, 2009); a high P_F value leads to a high degree of preferential flow (van Schaik, 2009; Bargues Tobella et al., 2014). It can be calculated as follows:

$$P_F = \left(1 - \frac{U_F \cdot W}{D_T}\right) \times 100\% \quad (1)$$

where W is the width of the corresponding soil profile ($W = 50$ cm), D_T is the total stained area of the soil profile (cm^2), and U_F is the depth at which the dye coverage decreases below 80 %, which assumed to be the infiltration front of the area dominated by uniform flow (van Schaik, 2009; Bargues Tobella et al., 2014).

2.4. Statistical analysis

One-way ANOVA was used to compare the variability between soil particle size distribution, soil bulk density, total porosity, capillary porosity, and natural water content, the density of different size classes of rock fragments, root biomass, root bulk density, and root length density for different grades of rocky desertification at a significance level

of 5 % by IBM SPSS 22.0 software (SPSS Inc., USA). The correlations between soil, rock fragments, and plant roots were analyzed using the Person correlation analysis by ggplot2 3.3.3 package in R (version 3.6.3).

3. Results

3.1. Soil physical properties of KRD

An analysis of the particle size distribution among all grades of rocky desertification shows that soil silt content accounts for the highest proportion of over 50 %, followed by clay content (16.65 %–18.42 %). The proportions of clay content are decreased with rocky desertification going severeness (Table 3). The higher the rock desertification grade is, the higher the sand content and the thinner surface soil. With the increase of rocky desertification, the soil bulk density increases, and the soil colour deepens. On the contrary, the hydrological characteristics parameters of the soil such as total porosity, capillary porosity, and natural volumetric water content gradually become lower (Table 3).

3.2. Effects of roots on preferential flow

90 % of the root biomass is distributed in the top 30 cm of the soil profile (Fig. 2a). Under different rocky desertification, a series of decreasing gradient of root biomass, root bulk density, and root length density is corroborated with soil layer going deep (Fig. 2). Sites with more severe grades of rocky desertification have low root biomass. The root biomass of the NRD and PRD is 3.50 and 2.39 times higher than SRD. The root biomass in the LRD, MRD, and SRD is primarily spread in the top 0–20 cm of the soil. Over 70 % of the total root volume and total root length are distributed in the 0–20 cm soil surface layer. The root bulk density of NRD is 8.79 %, 31.04 %, 57.35 %, and 93.68 % of those of PRD, LRD, MRD, and SRD, respectively. Regarding root length density, NRD is 20.44 %, 19.49 %, 47.72 %, and 73.28 % of those of PRD, LRD, MRD, and SRD, respectively.

Overall, P_F tends to increase and then decrease with increasing root biomass, root bulk density, and root length density (Fig. 3). Particularly, the root bulk density (adj. $R^2 = 0.824$, $n = 30$) and root length density (adj. $R^2 = 0.822$, $n = 30$) show high correlation coefficients with P_F . As the grade of rocky desertification increases, the values of the root system indicators decrease, but the trend in preferential flow is divided into two segments, which shows that other factors besides the roots also affect the preferential flow.

3.3. Effects of rock fragments on preferential flow

Whether surface outcrops or subsurface rock fragments, the density and volume fraction increase with the degree of rocky desertification (Fig. 4). The diameter of outcrops in the sites with severe rocky desertification was generally larger. The outcrops of PRD were mainly cobbles, and the outcrops of LRD and MRD were dominated by stones, while boulders larger than 600 mm were dominant in SRD (Fig. 4a). Rock fragments in soils were mainly in the diameter range of 2–600 mm, and rock fragments larger than 600 mm were primarily distributed in the MRD and SRD sites (Fig. 4b–c). A few small-sized rock fragments were found in the soils of NRD.

The expansion of the rock fragment coverage is the proportion of rock fragments in two-dimensional soil profile, which reflect the degree of underground rocky desertification to some extent. The dye-stained soil profile shows that water flow in stony soils percolates along the soil-rock interface compared to NRD (Fig. 5). The microenvironment at the soil-rock interface provides partial preferential paths for water infiltration. The microenvironment, the dynamic interphase between the rock fragments and the surrounding crushed soil matrix, is a region of high macroporosity. The interphase is not completely filled with fine soil matrix due to the obstruction of rock fragments (Nasri et al., 2015).

Table 3
Basic physical characteristics of soils.

Rocky desertification degree	Soil particle size distribution (%)			Soil bulk density (g/cm ³)	Total porosity (%)	Capillary porosity (%)	Natural water content (%)	Surface soil thickness (cm)	Soil types
	Clay content < 0.002 mm	Silt content 0.002–0.05 mm	Sand content 0.05–2 mm						
NRD	18.36 ± 1.52a	73.25 ± 2.31a	8.39 ± 1.20c	1.008 ± 0.05b	60.00 ± 0.82a	53.89 ± 1.24a	25.94 ± 2.68a	60–70	Red soil
PRD	18.42 ± 2.31b	72.02 ± 2.02b	9.56 ± 1.06bc	1.036 ± 0.12b	59.48 ± 0.69a	53.37 ± 1.08a	20.36 ± 2.42b	35–40	Red soil, limestone soil
LRD	17.93 ± 0.89b	68.68 ± 2.32bc	13.39 ± 1.18bc	1.065 ± 0.06b	55.02 ± 1.29b	50.74 ± 0.68b	17.31 ± 0.97c	25–38	Red soil, limestone soil
MRD	17.55 ± 1.05c	67.47 ± 3.12c	14.98 ± 1.65bc	1.158 ± 0.14ab	52.63 ± 1.11c	47.82 ± 0.59 cc	15.47 ± 1.45c	20–30	Grey limestone soil
SRD	16.65 ± 1.23d	69.13 ± 2.45d	14.22 ± 1.74a	1.256 ± 0.13a	49.53 ± 1.16d	50.16 ± 0.93d	10.36 ± 0.61d	10–18	Black limestone soil

Note. Textural class is based on USDA soil taxonomy classification. ± represent standard deviations. Lowercase letters indicate significant differences of volume fraction between different grades of rocky desertification ($p < 0.05$).

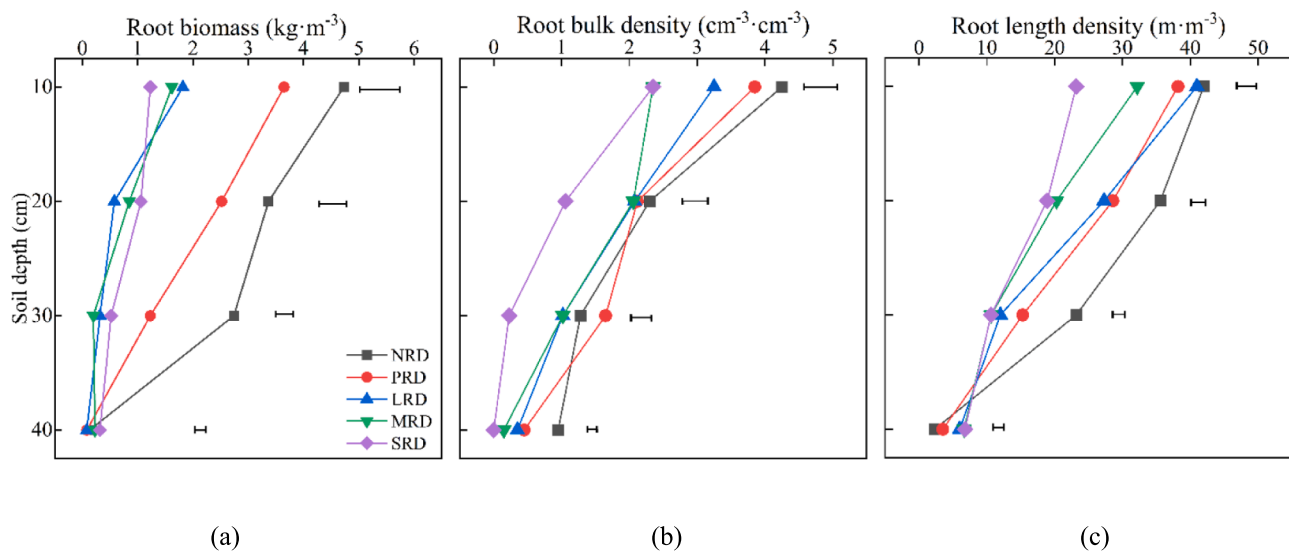


Fig. 2. Variation of (a) root biomass, (b) and (c) root length density with soil depth under different rocky desertification sites. The horizontal bars represent least significant difference at $p < 0.05$ for comparison under different rocky desertification at each depth increment.

This microenvironment is caused by physical, chemical, and ecological processes between the soil and rock fragments (Beven and Germann, 1982).

P_F obviously shows a trend of increasing and then decreasing under different rocky desertification (Fig. 6). PRD and LRD promoted the development of preferential flow, while MRD and SRD inhibited it compared with NRD (Fig. 6b). P_F ranges from 42.38 % to 56.95 % initially with the increase in rock fragment, but decreased to 39.41 % with further increasing in rock fragments (Fig. 6). With an increased in rock fragments, soil matrix dropped, which is most likely regions of limited flow mobility as evidenced by Fig. 6. As well, P_F –AVF fitted curve indicates that the P_F is highest when AVF increases to approximately 30 % ($R^2 = 0.911$, Fig. 6a).

3.4. Combined effects of soil – root – rock fragment system

According to Pearson correlation analysis (Fig. 7), the correlations between soil parameters, root system indicators, and rock fragment indices are apparently different in the study sites under different grades of rocky desertification. In NRD, clay and silt content show positive relationships with RD, RV, RB, RBD, and RLD (Fig. 7a). However, this is not entirely the case in rocky desertification sites, where clay and silt

content show negative correlations or no correlation with some of the root system indicators and rock fragment indices (Fig. 7b–e). The intricate interrelationships among the factors within the rocky desertification sites are illustrated indirectly.

Two PCA of 22 (i.e., 7 soil property variables, 12 rock fragment variables, and 3 root variables) variables are performed to evaluate the factors influencing preferential flow at different grades of rocky desertification (Fig. 8). As shown in Fig. 8a, the first two principal components explain 77.7 % (PC1) and 12.3 % (PC2) of the total variance, respectively, with cumulative explanatory contribution rates of 90 %. Therefore, two principal components are used as composite variables to evaluate the influencing factors of preferential flow under rocky desertification conditions. The PC1 is highly positively correlated with rock fragment volume fraction for particle sizes 7–26 mm (0.93) and 250–600 mm (0.93), and negatively correlated with the natural water content (–0.94) as well as total porosity (–0.94). PC2 is highly positively correlated with outcrop density for particle sizes 7–26 mm (0.87) and root bulk density (0.81). Therefore, PC1 contributes more to the rock fragment volume properties and soil water content characteristics, while PC2 mainly contributes to the outcrop density and root characteristics. MRD and SRD have a stronger relationship with PC1 positively, while NRD and PRD have a stronger relationship with PC1 negatively

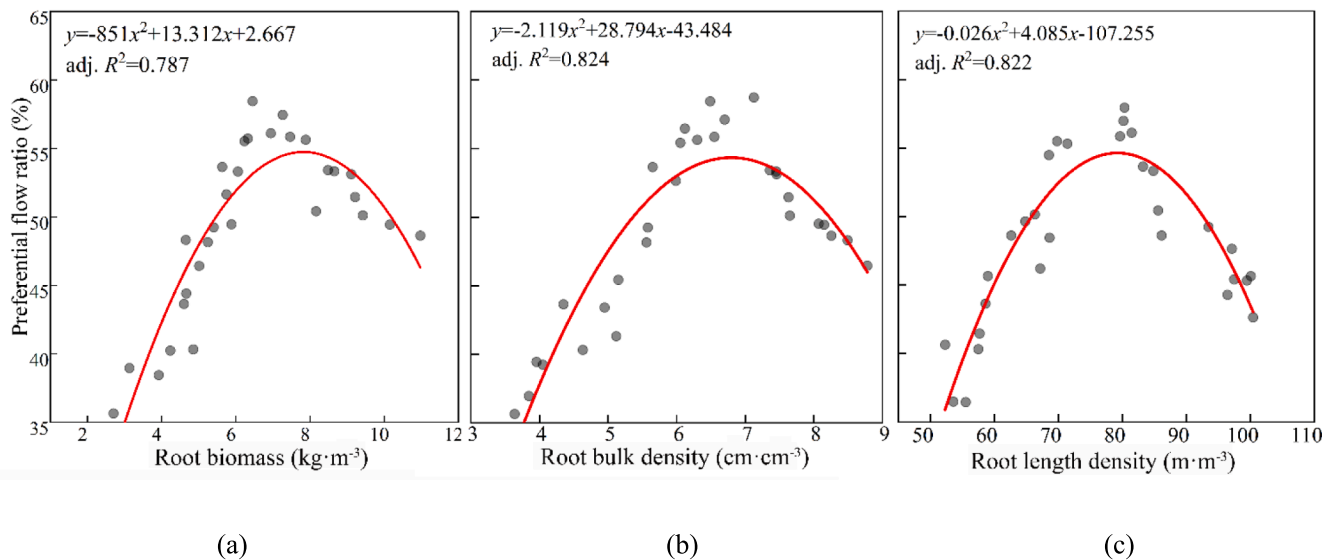


Fig. 3. Relation between preferential flow ratio (P_F) and root (a) root biomass, (b) root bulk density, and (c) root length density in all experimental plots. $n = 30$ for the P_F in all experimental plots.

(Fig. 8b). There is a greater relationship between LRD and PC2 positively (Fig. 8b). The five rocky desertification plots were clearly separated on PC1 (Fig. 8b). NRD and SRD were distinguished and influenced by root system characteristics and rock fragment density, respectively.

4. Discussion

4.1. Effects of soil on preferential flow

The differences in soil property, rock fragment distribution, and root trait in karst areas form different geomorphic landscapes. The prevalent soil type in rocky desertification areas is silt-clay; however, with the increasing rocky desertification grade, the proportion of sand shows a higher gradient. Limestone tends to exhibit strong karstification, with very intense hydro-chemical reactions at the rock-soil interface, and soils often contain dissolved species and sand-sized weathered rock fragments, resulting in high sand content (Peng et al., 2020). These coarse ossified particles interact with each other, and they are screened and rearranged (Tang et al., 2016). Therefore, soil bulk density is higher while soil porosity is lower in MRD and SRD. Changes in rock fragment content affect plant growth by influencing the soil structure. Consequently, hillslopes with high degrees of rocky desertification tend to be disadvantaged in terms of soil water and fertilizer retention, as well as the resistance of plant roots to growth increases (Dexter, 1988; Huang et al., 2023).

Our study indicated that soil physical properties, rock fragment distributions, and root characteristics all have an impact on the preferential flow pathways in soil – root – rock fragment systems in shallow karst. Soil properties and grain size composition are the main causes of slope erosion (Liu et al., 2023). Wang et al. (2023) considered that soil bulk density, clay, sand, silt contents, and rock fragments at the hillslope scale could influence soil water potential and preferential flow. In addition, soil porosity also has a very high response to soil moisture content. Soil water content with high soil porosity increases rapidly after rainfall. Soil porosity also determines soil permeability and has important effects on soil moisture conservation (Luo et al., 2023).

Studies have shown that percentage of mineral particles (sand, silt, clay) has the highest correlation with soil erodibility (Luo et al., 2022b). In KRD areas, the soil content is dominated by silt reaching more than 60 %, followed by clay, although higher rocky desertification grades in our study area led to higher soil sand content and lower clay content. Luo et al., (2022b) considered that silt is the major reason for

vulnerability of the soil to water erosion. In this research, the trend in soil content of rocky desertification grades is similar to Peng et al. (2020). Soils with high clay content tend to form fissures and pores (Waller and Wallender, 1993); in addition, laterite soils developed from limestone have a special dilation-shrinkage nature that exacerbates drought and water leakage.

4.2. Effects of roots on preferential flow

The intensification of rocky desertification has caused a decrease in vegetation cover. Therefore, root biomass, root bulk density, and root length density have decreased as a result of the succession of plant communities from higher to lower levels (Feng et al., 2016; Yu et al., 2022). Meanwhile, the interpenetration and network consolidation of plant roots can change the soil permeability and porosity. Rocky desertification areas have special soil-root-root structures and characteristics, leading to a vicious circle of poor and low nutrient earth, weak anti-disturbance ability and poor vegetation restoration effect (Maier et al., 2019; Huang et al., 2023).

In rocky desertification areas, water can be transported along plants' root channels, which is the key to plants' ability to withstand drought (Zhang et al., 2016). The root system achieves faster penetration mainly by producing macropores as preferential channels (Guo et al., 2019). Zhang et al. (2016) considered that the root length density that produces the preferential flow pathways is at least 50 % of the total root length density. The fine root not only grows into preferential flow channels but also creates preferential flow biopores, cracks, and burrows (Noguchi et al., 1997).

Our results find that an increasing degree of preferential flow is firstly displayed with increasing root indices. However, it is worth noting that the degree of preferential flow does not increase indefinitely with the increasing of the root system indicators, but shows a downward trend when the root system indicators increase to a certain level. These findings are not completely consistent with those of Zhang et al. (2016), who considered that positive correlations between fine plant roots and changes in the soil water content in northern China. One reason for this phenomenon is that the soil of northern China is predominantly sand, while the soil of KRD areas is silt and clay. The mechanical composition and compactness of soil affect soil macropores, which is the direct path of water infiltration downwards. On the other hand, rock fragment, as a dominant factor of widespread distribution, will also have a positive or negative impact on preferential pathways. Therefore, the degree of

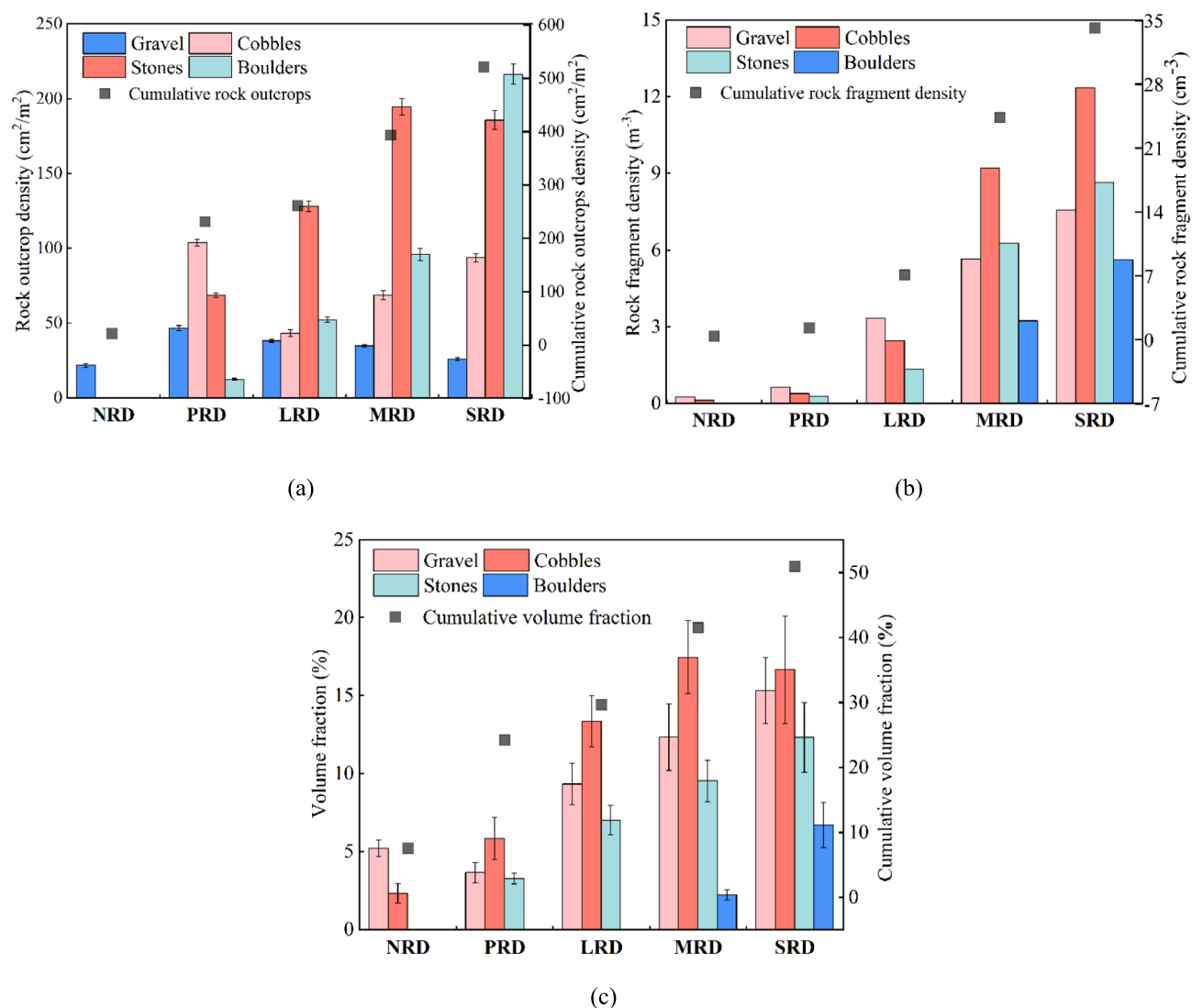


Fig. 4. Rock outcrop density (a), subsurface rock fragment density (b), and volume fraction (c) of study sites with different grades of rocky desertification and cumulative amount of each degree.

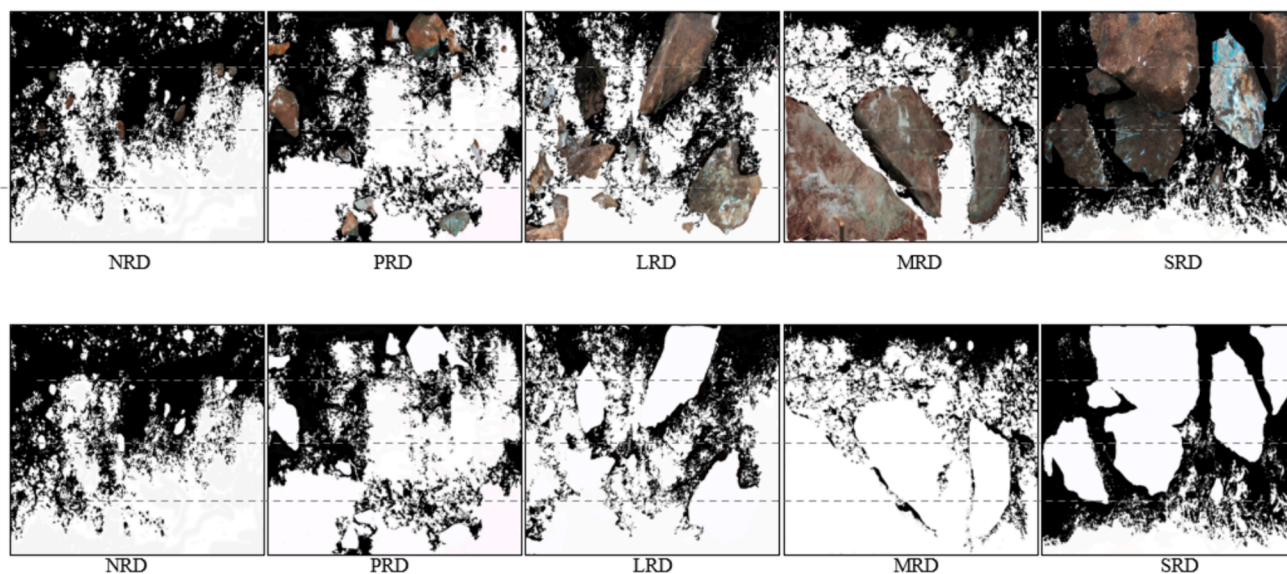


Fig. 5. Dye-stained soil profiles with and without rock fragments.

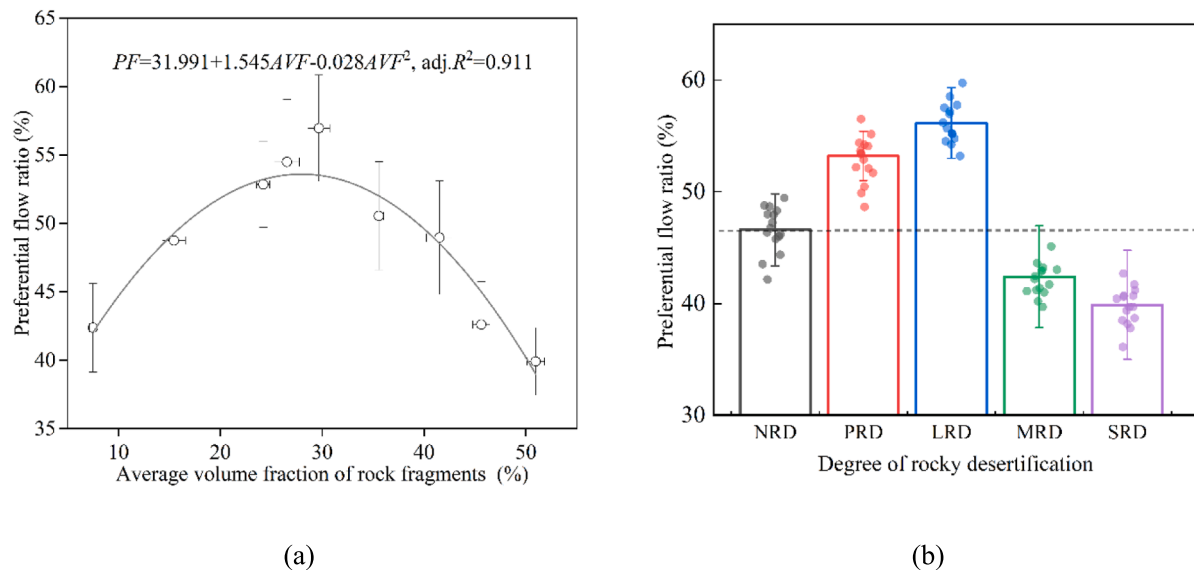


Fig. 6. Preferential flow ratio (b) as a function of average volume fraction (PF, preferential flow ratio, %; AVF, average volume fraction of rock fragment, %) and (a) under different grades of rocky desertification.

preferential flow is not only controlled by root systems, but the influence of rock fragment also should be taken into account (Fig. 9). Wu et al., (2021a) believed that preferential flow and roots are mutually reinforcing, with preferential flow promoting root growth and further improving soil nutrients.

Some studies suggest that preferential pathways exchange along the plant root system is more visible than that between the rock surface and soil matrix (Backnäs et al., 2012). Mair et al. (2022) suggested that preferential flow occurred according to the direction and distribution of roots in the soil, and he used modified Richards' equation to derive the hydraulic properties of the vegetated soil as a result of root-oriented preferential flow.

4.3. Effects of rock fragments on preferential flow

Studies have shown that the pooling of rainwater by outcrops leads to a higher soil bulk density at the soil-rock interface under long-term erosion (Zhao et al., 2018). This also explains the increasing in the grade of rocky desertification leading to the increasing in soil bulk density. The weathering of carbonate rock fragments occurs mainly at the earth interface, where the surface soil accumulates between the rock fragments. Rock fragments in soils have essential effects on soils, soil moisture, plant growth, fertility, and overflow hydraulics, as well as on the erosion of hillsides and inter-hills (Poesen and Lavee, 1994). The position of the rock fragments, on the surface or inside the soil profile, free on the surface or embedded in the fine earth, has a large influence on the total water properties of the soil (Zhang et al., 2011), especially because the rock fragments alter evaporative conditions at the soil surface (Perez, 1998). Chen et al. (2011) considers that the dominant factors affecting the spatial distribution of rock fragments may be related to slope gradient, vegetation and geomorphologic condition, to be followed by further research.

The different grades of rocky desertification are essentially due to the weathering of the bedrock, which affects the distribution of soil grain-size and rock fragment composition (Wu et al., 2021b). Although the results pointed that surface outcrop density, subsurface rock fragment density, and rock fragment volume fraction increased with the degree of rocky desertification, the degree of preferential flow did not coincide with the trend in rock fragments. Surface outcrops have no direct effect on groundwater infiltration and preferential flow; however, Zavala et al. (2010) reported that increased soil surface gravel leads to increased

infiltration. Rock fragments resting on (or partially embedded in) the topsoil not only retarded surface flow and dissipated energy by increasing surface roughness, but also protected the soil surface from erosion by reducing the splash effect of raindrops (Zhang et al., 2016).

The LRD exhibited the highest degree of preferential flow development (PF: 56.9 %) with a rock fragment volume fraction of 16.2 %. The PF of LRD was 1.4 times that of NRD and 1.1–1.3 times that of other grades of rocky desertification. These results suggested that rock fragments can promote preferential flow but there is a threshold. A similar finding was obtained by Hou et al. (2023), who considered that rock fragments reduced the main paths of preferential flow or had no effect when the total volume fraction of rock fragments exceeds 30 %. Zhang et al. (2016) identified that the rock fragment area is regarded as an area of limited flow mobility; however, as the rock fragment content increased further, a continuous preferential flow path or interphase formed between rock fragments and the surrounding soil matrix. The content of subsurface rock fragments determines the volume, effective size, and connectivity of the macropores space, and several related studies have noted the influence of rock fragments on subsurface erosion (Naseri et al., 2019; Liu et al., 2019).

4.4. Preferential flow responses to soil – root – rock fragment system

For a long time, the response of preferential flow to soil – root – rock fragment systems and their impact on subsurface hydrological processes have been neglected in rocky desertification areas. The development of preferential flow is not only the result of the unilateral action of rock fragments or roots. In severe rocky desertification areas, poor root systems caused by sparse vegetation and the large soil matrix occupied by rock fragments instead inhibited the rapid infiltration of effective water. Rock fragment is the main limiting factor inhibiting MRD and SRD moisture movement (Fig. 9). For MRD and SRD hillslopes, natural restoration should be the mainstay of revegetation to avoid water leakage exacerbated by disturbance of the soil – root – rock fragment systems (Yu et al., 2023b). Our study concluded that the potential and light rocky desertification areas have stronger degree of preferential flow, which implied that groundwater and nutrient leakage is a serious problem in these areas. The limitations of PRD and LRD moisture movement are soil – root – rock fragment systems and root systems, respectively (Fig. 9). For these hillslopes where the degree of rocky desertification is not yet fully matured, the vegetation cover should be



Fig. 7. Pearson correlation coefficients between soil parameters, root system indicators, and rock fragment indices (a–e) and correlation of preferential flow ratio with variables (f). Clay%: clay content; Silt%: silt content; Sand%: sand content; SBD: soil bulk density; TP: total porosity; CP: capillary porosity; NWC: natural water content; OD(2–76): outcrop density in the diameter range of 2–76 mm; OD(76–250): outcrop density in the diameter range of 76–250 mm; OD(>600): outcrop density in the diameter range higher than 600 mm; RD(2–76): rock fragment density in the diameter range of 2–76 mm; RD(76–250): rock fragment density in the diameter range of 76–250 mm; RD(>600): rock fragment density in the diameter range higher than 600 mm; RV(2–76): rock fragment volume in the diameter range of 2–76 mm; RV(76–250): rock fragment volume in the diameter range of 76–250 mm; RV(>600): rock fragment volume in the diameter range higher than 600 mm; RB: root biomass; RBD: root bulk density; RLD: root length density; NRD-PF: preferential flow ratio for non-rocky desertification; PRD-PF: preferential flow ratio for potential rocky desertification; LRD-PF: preferential flow ratio for light rocky desertification; MRD-PF: preferential flow ratio for moderate rocky desertification; SRD-PF: preferential flow ratio for severe rocky desertification. * $p < 0.05$, ** $p < 0.01$.

increased (Guo et al., 2023; Zhao et al., 2018). Water- and soil-retaining plants are planted to form a perennial network of roots to close pore spaces and holes, thus preventing surface and subsurface soil loss (Zhang et al., 2010; Guo et al., 2022).

Therefore, appropriate management measures should be adopted for rocky desertification hillslopes in the light of different ecological

environments in order to enhance the effectiveness of rocky desertification management. Because of the large vertical differentiation of rocky hillslopes, future research should be carried out in accordance with the characteristics of the vertical zoning of rocky hillslopes, so as to carry out the prevention and control of rocky desertification in accordance with local conditions.

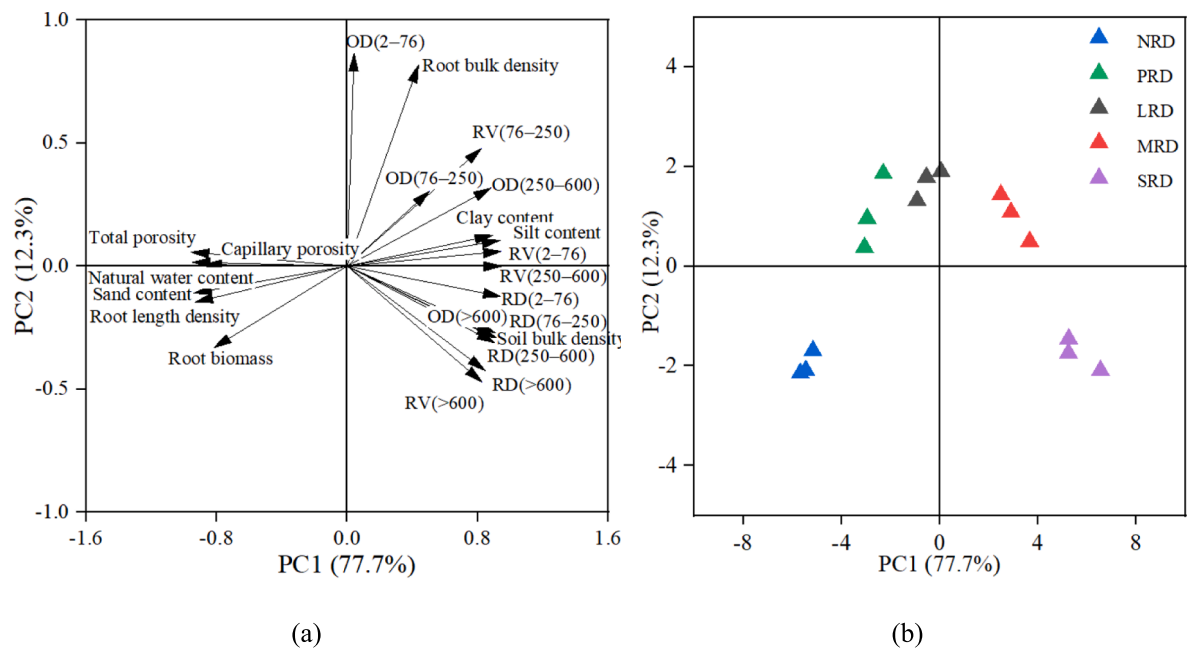


Fig. 8. Two-dimensional (2D) principal component analysis (PCA) plots. (a) Loading plot and (b) score plot for impact variables of preferential flow. NRD: Non-rocky desertification; PRD: Potential rocky desertification; LRD: Light rocky desertification; MRD: Moderate rocky desertification; SRD: Severe rocky desertification; OD: Outcrop density; RD: Rock fragment density; RV: Rock fragment volume. The numbers in parentheses represent the particle size range.

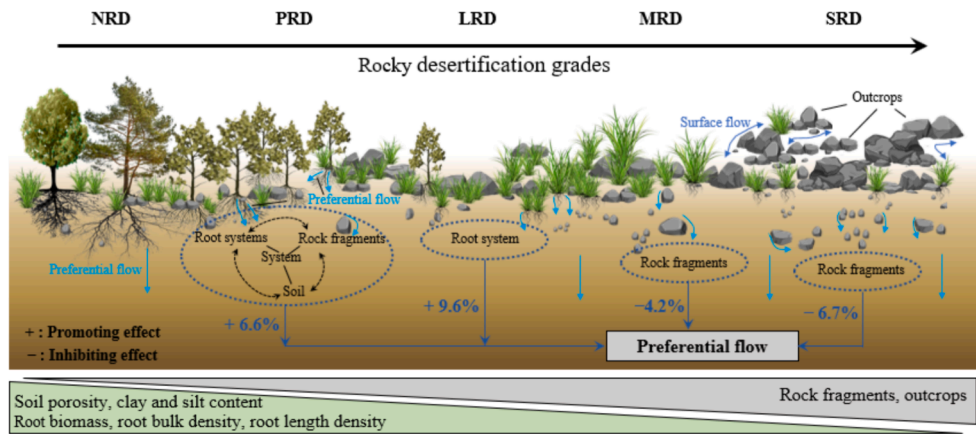


Fig. 9. Schematic diagram of the response of different grades rocky desertification to preferential flow.

5. Conclusions

In this study, we examined the characteristics of preferential flow changes for five grades of rocky desertification, i.e., NRD, PRD, LRD, MRD, and SRD. The study focused on the response of preferential flow to soil properties, rock fragments, and plant roots, providing a reference for water loss and land management in rocky desertification ecosystems.

In order to promote the efficient management of ecosystems in rocky desertification areas, the focus should be on the ecological environment of potential and light rocky desertification areas. These areas have a higher development of preferential flow and may have potential soil erosion problems. At such stages, which have not yet reached ecosystem irreversibility, appropriate plant and rock fragment ratios can effectively avoid water loss in epikarst. Our research provides a new perspective for regional vegetation restoration and regulation in the process of rocky desertification control.

CRediT authorship contribution statement

Fang Hou: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Jinhua Cheng:** Writing – review & editing, Funding acquisition. **Han Zhang:** Investigation. **Xianglian Wang:** Investigation. **Dewei Shi:** Investigation. **Ning Guan:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors gratefully acknowledge the editor and reviewers. This research was supported by the National Natural Science Foundation of China (no. 32071839), National Science and Technology Support Project (2018ZX07101005), and National Natural Science Foundation of Joint Fund for Changjiang River Water Science Research (U2340215).

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